

I used option 1: [Tutorial: Sentiment Analysis in R](#)

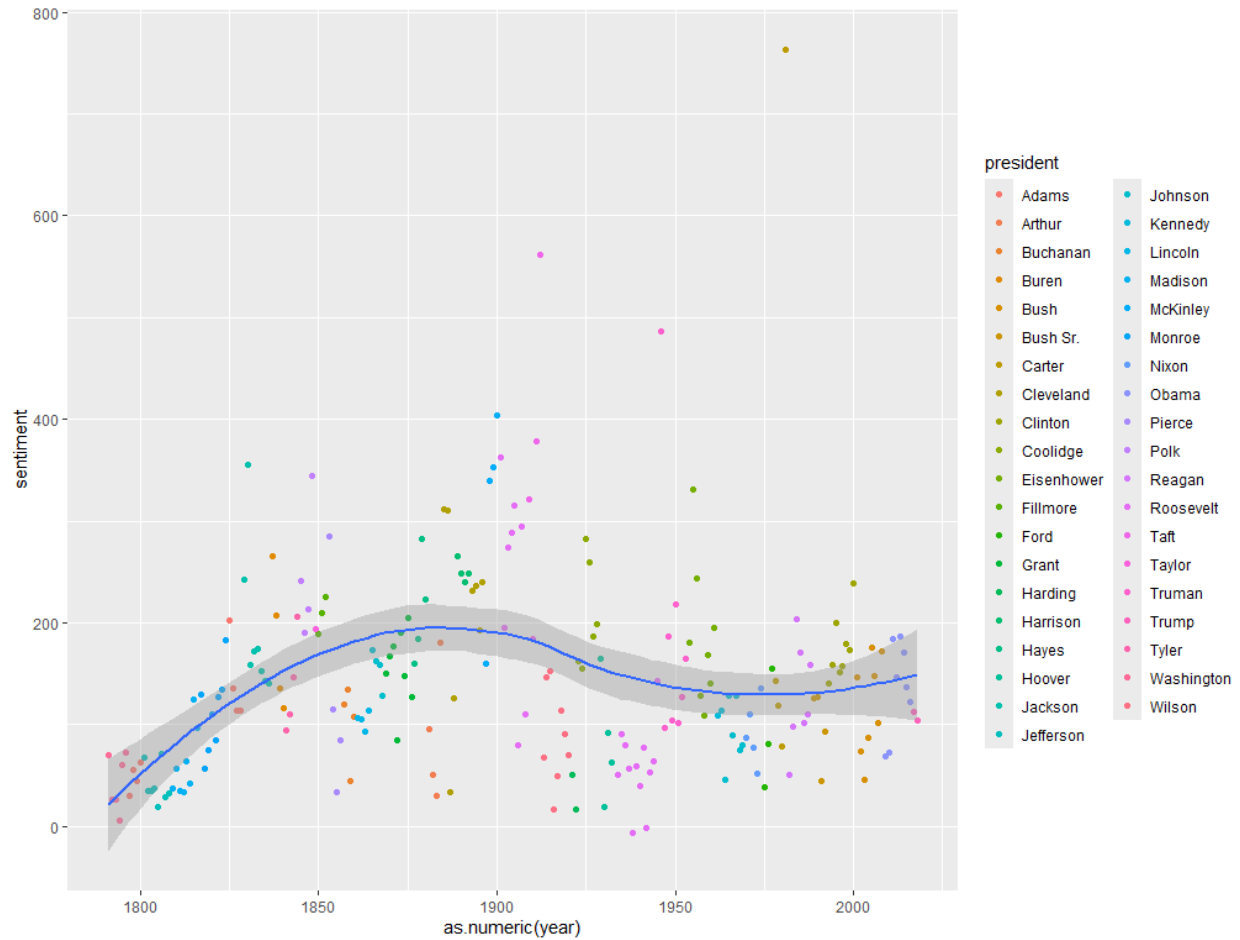
```
> # get the sentiment from the first text:
> tokens %>%
+   inner_join(get_sentiments("bing")) %>% # pull out only sentiment words
+   count(sentiment) %>% # count the # of positive & negative words
+   spread(sentiment, n, fill = 0) %>% # made data wide rather than narrow
+   mutate(sentiment = positive - negative) # # of positive words - # of negative words
Joining with `by = join_by(word)`
# A tibble: 1 x 3
  negative positive sentiment
  <dbl>      <dbl>      <dbl>
1      59       89        30
```

Interpretation: Adams_1797.txt has 59 negative words, 89 positive words, and a sentiment score of 30. This indicates the speech was overall positive, since positive words outnumbered negative words.

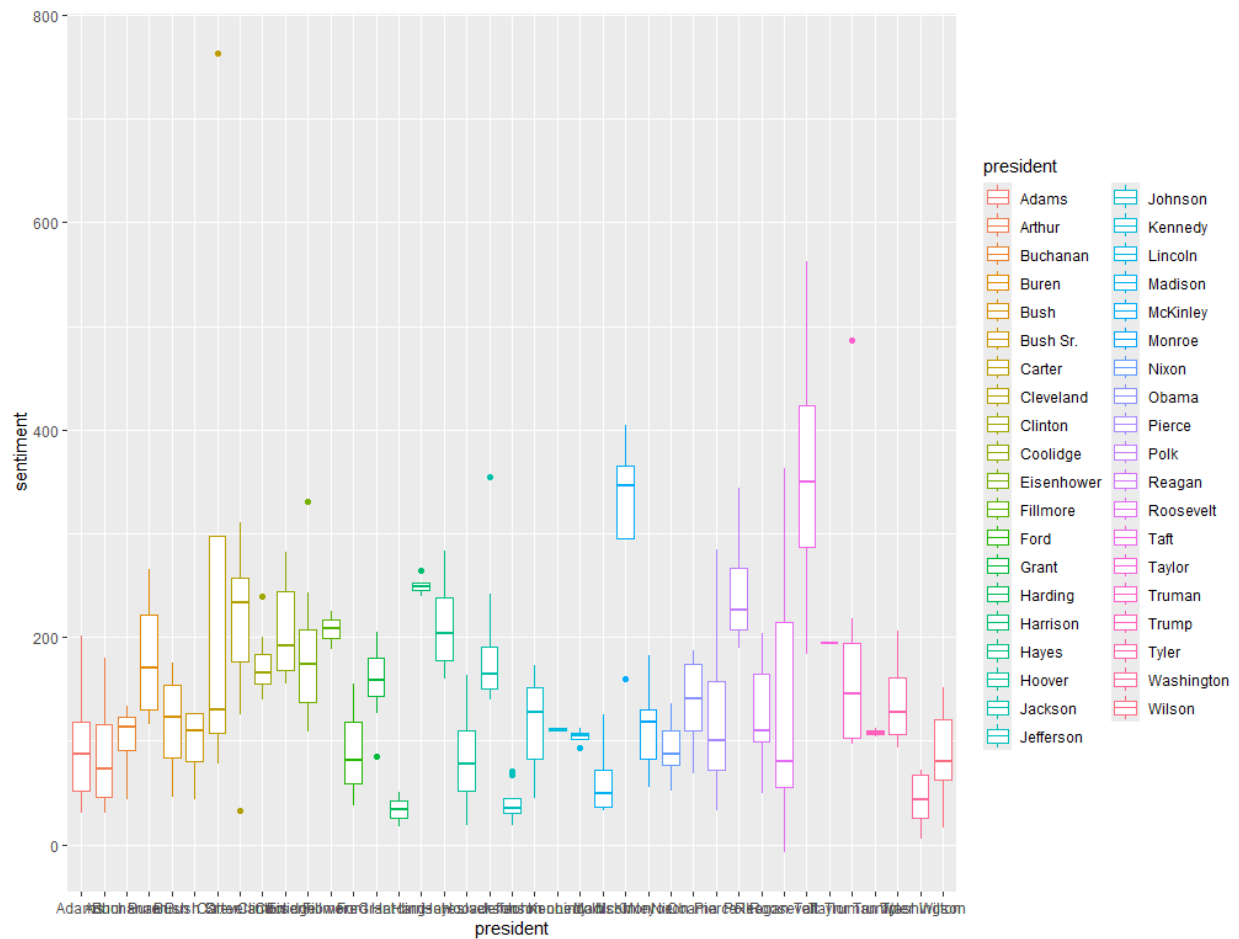
```
> # summarize the sentiment measures
> summary(sentiments)
```

	negative	positive	sentiment	file	yea
r					
Min.	: 20.0	Min. : 57.0	Min. : -7.0	Length:227	Min. :
1791					
1st Qu.:	101.0	1st Qu.: 180.5	1st Qu.: 76.0	Class :character	1st Qu.:
1848					
Median :	138.0	Median : 282.0	Median :128.0	Mode :character	Median :
1904					
Mean :	177.5	Mean : 321.3	Mean :143.8		Mean :
1904					
3rd Qu.:	209.5	3rd Qu.: 386.5	3rd Qu.:184.0		3rd Qu.:
1962					
Max.	:820.0	Max. :1458.0	Max. :763.0		Max. :
2018					
president					
Length:227					
Class :character					
Mode :character					

Interpretation: Most speeches are positively skewed. The average speech contains more positive than negative words. There are some extremely high sentiment values.



Interpretation: Sentiment increases during the 1800s, there is high variability in the early 1900s, sentiment dips around mid-1900s, and there is moderate stabilization in modern years. Overall, presidential tone fluctuates over time rather than steadily increasing.



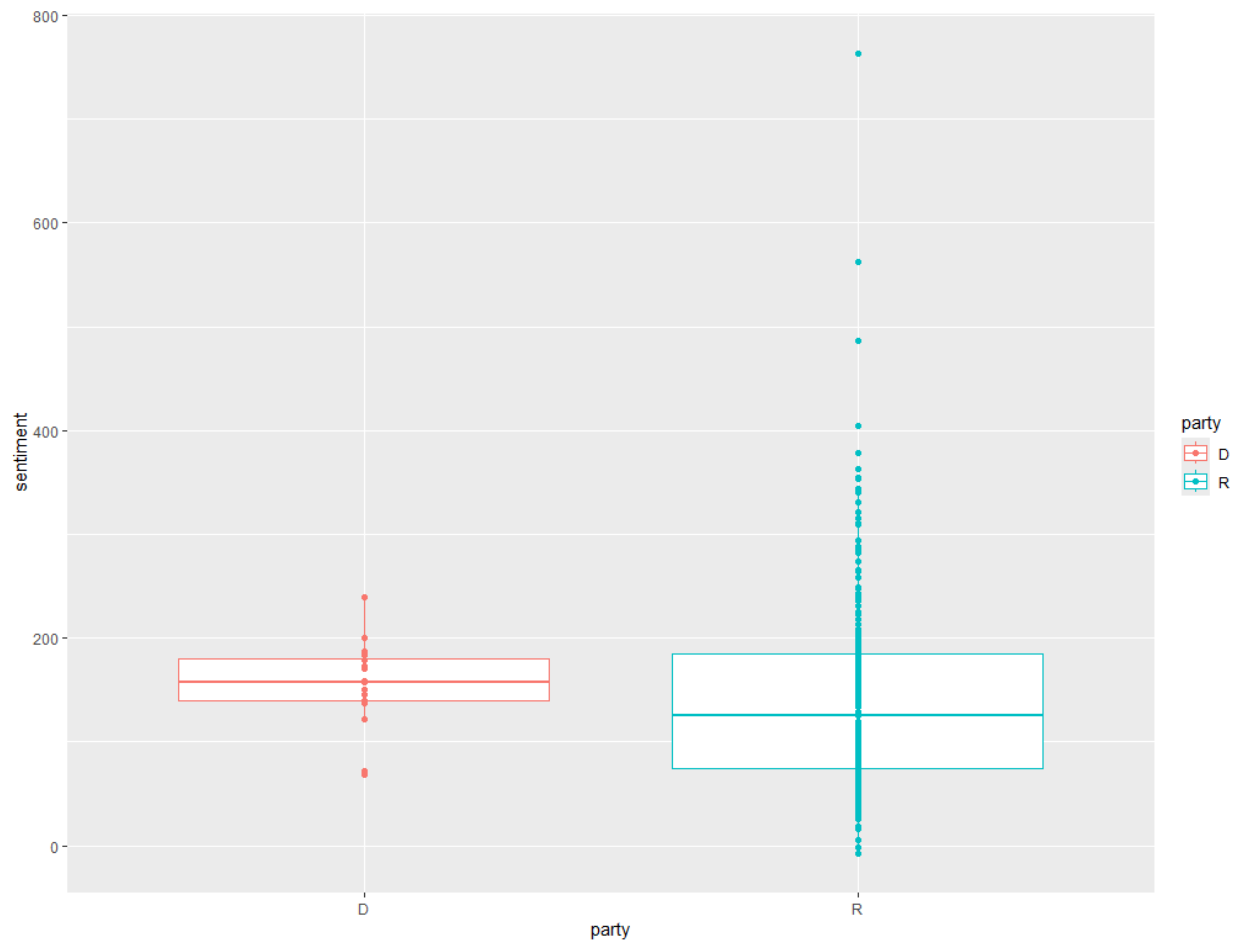
Interpretation: Some presidents, like Reagan and Roosevelt, show very high variability, others have consistently moderate sentiment, and several presidents show extreme positive outliers. Overall, presidential tone varies more within presidents than across presidents.

```
> # the difference between the parties is significant  
> t.test(democrats$sentiment, republicans$sentiment)
```

```
Welch Two Sample t-test
```

```
data: democrats$sentiment and republicans$sentiment  
t = 0.96626, df = 29.756, p-value = 0.3417  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
-13.87927  38.79041  
sample estimates:  
mean of x mean of y  
155.3750  142.9194
```

Interpretation: Democrats have slightly higher average sentiment (155 vs 143). However, $p = 0.3417 > 0.05$, so we fail to reject the null hypothesis. There is no statistically significant difference in sentiment between Democratic and Republican presidents.



Interpretation: The distributions overlap heavily, republicans show slightly greater variability, democrats have a slightly higher median, but overlap confirms the t-test result. Overall, party affiliation does not meaningfully explain variation in sentiment tone in State of the Union addresses.

Full code:

```

1 # load in the libraries we'll need
2 library(tidyverse)
3 library(tidytext)
4 library(glue)
5 library(stringr)
6 library(dplyr)
7 library(tidyr)
8
9 # get a list of the files in the input directory
10 input_folder <- "c:/Users/carol/OneDrive/Desktop/DATA 420/State of the Union Corpus (1790 - 2018)/sotu"
11 files <- list.files(input_folder)
12
13
14
15 # stick together the path to the file & 1st file name
16 fileName <- glue("{input_folder}/{files[1]}")
17 # get rid of any sneaky trailing spaces
18 fileName <- trimws(fileName)
19
20 # read in the new file
21 fileText <- glue(read_file(fileName))
22 # remove any dollar signs (they're special characters in R)
23 fileText <- gsub("\\$", "", fileText)
24
25
26 # tokenize
27 # got a data_frame() warning so fixed the code
28 tokens <- tibble(text = fileText) %>% unnest_tokens(word, text)
29 sentiments <- tibble()
30
31 # get the sentiment from the first text:
32 tokens %>%
33   inner_join(get_sentiments("bing")) %>% # pull out only sentiment words
34   count(sentiment) %>% # count the # of positive & negative words
35   spread(sentiment, n, fill = 0) %>% # made data wide rather than narrow
36   mutate(sentiment = positive - negative) # # of positive words - # of negative words
37
38 # write a function that takes the name of a file and returns the # of positive
39 # sentiment words, negative sentiment words, the difference & the normalized difference
40 Getsentiment <- function(file) {
41   fileName <- file.path(input_folder, file)
42
43   fileText <- read_file(fileName) %>%
44     gsub("\\$", "", .) %>%
45     gsub("[^[:alnum:]]\\s+", " ", .) %>%
46     gsub("\\s+", " ", .)
47
48   tokens <- tibble(text = fileText) %>%
49     unnest_tokens(word, text) %>%
50     mutate(word = str_to_lower(word))
51
52   sentiment_counts <- tokens %>%
53     inner_join(get_sentiments("bing"), by = "word", relationship = "many-to-many") %>%
54     count(sentiment) %>%
55     pivot_wider(names_from = sentiment, values_from = n, values_fill = 0)
56
57   if (!"positive" %in% colnames(sentiment_counts)) sentiment_counts$positive <- 0
58   if (!"negative" %in% colnames(sentiment_counts)) sentiment_counts$negative <- 0
59
60   sentiment_counts %>%
61     mutate(
62       sentiment = positive - negative,
63       file = file,
64       year = as.numeric(str_extract(file, "\\d{4}")),
65       president = str_match(file, "(.*?)_")[2]
66     )
67 }
68
69 # test: should return
70 # negative positive sentiment file year president
71 # 117 240 123 Bush_1989.txt 1989 Bush
72 # Note** it's returning Adams_1797 because it is the first .txt in folder
73 Getsentiment(files[1])
74

```

```

75 # file to put our output in
76 sentiments <- data_frame()
77
78 # get the sentiments for each file in our dataset
79 for(i in files){
80   sentiments <- rbind(sentiments, GetSentiment(i))
81 }
82
83 # disambiguate Bush Sr. and George W. Bush
84 # correct president in applicable rows
85 bushSr <- sentiments %>%
86   filter(president == "Bush") %>% # get rows where the president is named "Bush"...
87   filter(year < 2000) %>% # ...and the year is before 200
88   mutate(president = "Bush Sr.") # and change "Bush" to "Bush Sr."
89
90 # remove incorrect rows
91 sentiments <- anti_join(sentiments, sentiments[president == "Bush" & sentiments$year < 2000, ])
92
93 # add corrected rows to data_frame
94 sentiments <- full_join(sentiments, bushSr)
95
96 # summarize the sentiment measures
97 summary(sentiments)
98
99 # plot of sentiment over time & automatically choose a method to model the change
00 ggplot(sentiments, aes(x = as.numeric(year), y = sentiment)) +
01   geom_point(aes(color = president)) + # add points to our plot, color-coded by president
02   geom_smooth(method = "auto") # pick a method & fit a model
03
04 # plot of sentiment by president
05 ggplot(sentiments, aes(x = president, y = sentiment, color = president)) +
06   geom_boxplot() # draw a boxplot for each president
07
08 # is the difference between parties significant?
09 # get democratic presidents & add party affiliation
10 democrats <- sentiments %>%
11   filter(president %in% c("Clinton", "Obama")) %>%
12   mutate(party = "D")
13
14 # get democratic presidents & party add affiliation
15 republicans <- sentiments %>%
16   filter(!president %in% c("Clinton", "Obama")) %>%
17   mutate(party = "R")
18
19 # join both
20 byParty <- full_join(democrats, republicans)
21
22 # the difference between the parties is significant
23 t.test(democrats$sentiment, republicans$sentiment)
24
25 # plot sentiment by party
26 ggplot(byParty, aes(x = party, y = sentiment, color = party)) + geom_boxplot() + geom_point()
27
28

```